Quotation No : Q24000587

Date : 22-Nov-2024

INGERSOLL-RAND (I) LTD.
Gurugram
GURGAON
HARYANA

Kind Attention : Mr./Ms. Sunil Khanduja / Mahesh Kumar Kr

Dear Sir,

We understand that in industry, reliability is not just a requirement,it is the foundation of success.
That is why Absstem Industrial PSA Generator is more than just a product; it is a commitment to excellence, efficiency, and uninterrupted performance.

Every component is designed with precision and innovation, ensuring that each molecule of oxygen and nitrogen is delivered with the utmost purity.

With fully automated operation, our system seamlessly integrates into your processes, reducing manual intervention and maximizing productivity.

Moreover, our industry-leading air ratio ensures that you achieve the highest efficiency, optimizing energy consumption while maintaining peak performance.
And because we believe in standing by our craftsmanship, we offer an unmatched-10 year warranty on sieves, giving you the peace of mind that your investment is protected for the long run.

We have attached the detailed quotation for your review.

Should you have any questions or require further customization to meet your specific needs, please do not hesitate to reach out.

We are here to provide not just a product but a solution that breathes life into your operations.

We look forward to the opportunity to serve you and build a lasting partnership.

Thanks and Regards,

For Absstem Industrial Solutions Pvt. Ltd.

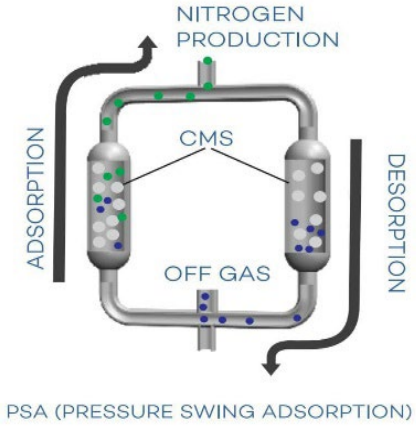
Amit Sahni
Chief Executive Officer
9582499594
www.absstem.com

nGEN POWERED BY ABSSTEM INDUSTRIAL

Industrial PSA Nitrogen Generator



PSA Technology



PSA (PRESSURE SWING ADSORPTION)

Industrial Nitrogen Generator mainly consists of two adsorbers filled with carbon molecular sieves. Pressurized air is passed through one of these adsorbers, carbon having the property to attract oxygen, adsorbs oxygen from air and Nitrogen is allowed to pass through and move into the storage tank as the final product. As the adsorber approaches saturation, the air is allowed to pass through the other adsorber while this adsorber is being regenerated to ensure a continuous supply of Nitrogen. This changeover takes place through a specially designed microcontroller to ensure a hassle-free operation. The quality of Nitrogen produced is monitored by a Nitrogen analyser.

Key Features

- Shipped ready for installation
- 5 minutes to run and never have to stop it!
- Easy installation
- Low operating expenses compared to gas cylinders/liquid Nitrogen
- Maintenance costs are very less
- Embedded Safety Interlocks
- Space-saving design
- Eliminates all hazards which are there with gas cylinder or liquid Nitrogen
- Longevity: at least 10 years

Absstem Industrial Solutions Pvt. Ltd.
Plot No. 591, Sector 8, Phase IV
IMT Manesar, Gurugram, Haryana - 122051
Contact - 1800 3010 3394 Email - indsales@absstem.com
CIN: U29256HR2022PTC103387

Description

a.	Product Name	Nitrogen Gas Generator
b.	Model Range	**NGEN**
	Technology	PSA (Pressure Swing Adsorption)

Specifications

a.	Nitrogen Flow Rate	As per Performance Sheet
b.	Nitrogen Purity	As per Performance Sheet
c.	Nitrogen Delivery Pressure	5.5 Barg (Higher Pressure Options Available)
d.	Ambient conditions	
	Temperature	0 to 50 Deg. Celsius
	Relative Humidity	up to 80 % (Max.)
	Altitude	up to 1000 m above Sea level

Scheme Offered – (PSA Only)

This scheme will comprise of

- Air package with inlet air filtration system
  (Air compressor, Air dryer, Water Separator, Oil filters)
- Industrial Nitrogen generator
- Storage Tank
- Nitrogen Analyzer
- Alarm panel

Unique Features of Absstem

a. All the fabricated items are cleaned and degreased following our strict quality policy.
b. All the Instruments are specially manufactured to make them suitable for Nitrogen gas.
c. Guaranteed 10+ years of healthy life.
d. We provide the purest **Nitrogen** in Industry.
e. Data logging and Graphical Reports for all parameters available.
f. Test certificates of all instruments are available on demand.

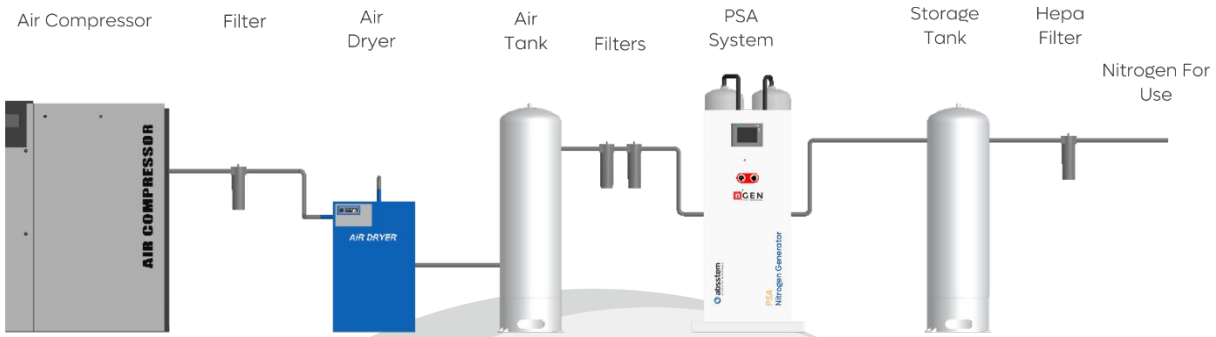
Onsite Gas Generation vs Liquid Tank/Cylinder Alternative

Non-stop availability: 24 hours a day, 7 days a week
Easy integration within existing installations
Your own independent supply of industrial gas
Significant economies for scale and lower operational costs: no rental charges, logistical expenses and bulk user evaporation losses
No safety hazards when handling high-pressure cylinders

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Equipment Description

The Industrial Nitrogen Generator consists of



Air Compressor	An Air compressor is used to compress the atmospheric air to a pressure required for the separation of **Nitrogen** from the air.
Refrigeration dryer	Compressed air then passes through a refrigerated dryer to remove moisture (which is harmful to sieves) from the air.
Air Tank	Dry air is collected in an air tank to remove the pulsations which serve the dynamic (changing) requirement of air by the PSA system.
Multistage Filtration system	Before air enters the PSA system, it is passed through multistage filtration removing moisture, particulates and oil content to the specified levels.
PSA system	Pressurized air is passed through PSA Towers filled with Molecular sieves to separate **Nitrogen** from Air.
Storage Tank	Industrial Grade **Nitrogen** produced from NGEN is stored in this tank to supply a continuous flow of **Nitrogen**.
Nitrogen Analyzer	Nitrogen produced is monitored by a **Nitrogen** analyser. In case, O2 purity goes below the specified level, it gives an alarm and switches it to vent mode.

HD Touch Screen Display

(User-friendly touchscreen interface for easy operation and maintenance)

Absstem industrial Nitrogen generator complies with all industry standards and regulation

- Alarm management system
- Plug and play
- Fully automatic
- Process data display
- Consumables running hours display
- Data logging via USB drive
- Graphical representation



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Additional upgrades

- Digital flow meter with totaliser
- High-pressure Nitrogen tank
- Cylinder manifold and alarm management system
- Remote monitoring

Plant Operation

- Unit operates 24-hrs a day, 7-days a week reliably, automatically & safely

Automation

- The Plant is fully automatic and needs only electrical power for running the Air compressor
- The machine automatically enters into standby mode when the level in the Storage Tank reaches its set limit. This helps to increase the life of the Air compressor.

International Quality standards

A strict quality procedure, following the leading international pharmacopoeia and Industrial directives, is followed in the selection of all mechanical and electrical instruments, vessel fabrication and designing of Industrial Nitrogen generators.

Itemised Supply Description

No.	Item description	Make	Qty
a	Air Compressor with Filters	ABSSTEM APPROVED	Quoted Separately
b	Refrigeration Dryer	ABSSTEM APPROVED	Quoted Separately
c	Air Tank	ABSSTEM	Quoted Separately
d	Set of Filters	ABSSTEM	1 no.
e	PSA unit (NGEN)	ABSSTEM	1 no.
f	Electrical Power Panel with PLC and all cabling	ABSSTEM & MITSUBISHI	1 no.
f	Digital Nitrogen Analyzer	ABSSTEM	1 no.
g	Storage Tank	ABSSTEM	Quoted Separately
h	Outlet product gas Filter	ABSSTEM	1 no.

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Electrical Protection

The client should make proper arrangements for earthing/grounding systems for the Industrial **Nitrogen** Generator at their facility.

Enclosure Panel

- Ergonomically designed metal enclosure with powder-coated finish for longer life.
- High-strength IP-rated panel fittings from Germany for a robust enclosure.
- Fan filters from **Rittal** are IP-55 rated making the electrical enclosure's internal environment dust-free.
- Easy connectivity to any place of installation with req. port connections.
- High-strength base frame designed for higher load and easy transportation.
- Vessels are designed per ASME section VIII DIV I. Vessels are epoxy paint coated which is considered good for longevity and corrosion resistance.
- All electrical items are CE certified with operating voltage at 24 VDC

Annual Maintenance Contract (AMC)

Our in-house support Absstem shield is 24*7 open and we assure a timely service maintenance schedule. Per the schedule mentioned in the operational manual, we suggest the operator change the filters by following a simple protocol.

It is very easy to change or replace as our commissioning engineer plus our commissioning engineer and tech support office will always guide you about the activity.

Best ROI Guaranteed

ROI within 6 to 18 months when compared to conventional methods

Generator performance depends on the component selection. Our processes make this job easier as all equipment is selected based on more than 3 decades of research. NGEN is a fully automatic (operator-free operation) that ensures peace of mind.

We deliver what we commit!!

Thank You

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Product Description

Product	Description	Make	Qty
nGen Nitrogen Generator	nGEN 1300 PC Nitrogen		1

Caption	Specification
Gas Generated	Nitrogen
Gas Pressure	8 Barg
Flow @ 99.99%	1306 Nm3/Hr
Input Voltage	230V
1 Phase	50 Hz
Input Power	0.3 KW
Module	Multi Tower PSA Module Only
Outlet Dew Point	(-)50 Degrees Celcius
Input Compressed Air Requirement	3900 CFM (FAD)
Input Pressure	9 Barg
Air Compressor Air Dryer and Air Vessel	In your Scope
Ambient Considered	40 Degrees C
Inlet Dew Point	3 Degrees Celsius PDP
Separation Media	Carbon Molecular Sieves and Mixbed of Adsorbants

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Product	Description	Make	Qty
nGen Nitrogen Generator	nGEN 1600 PC Nitrogen		1

Caption	Specification
Gas Generated	Nitrogen
Gas Pressure	8 Barg
Flow @ 99.99%	1568 Nm3/Hr
Input Voltage	230V
1 Phase	50 Hz
Input Power	0.3 KW
Module	Multi Tower PSA Module Only
Outlet Dew Point	(-)50 Degrees Celcius
Input Compressed Air Requirement	4680 CFM (FAD)
Input Pressure	9 Barg
Air Compressor Air Dryer and Air Vessel	In your Scope
Ambient Considered	40 Degrees C
Inlet Dew Point	3 Degrees Celsius PDP
Separation Media	Carbon Molecular Sieves and Mixbed of Adsorbants

Product	Description	Make	Qty
nGen Nitrogen Generator	nGEN 150 P Nitrogen		1

Caption	Specification
Gas Generated	Nitrogen
Gas Pressure	upto 6.5 Barg(Higher options Available)
Flow @ 99.99 %	367 Nm3/hr
Flow @ 99.995 %	278 Nm3/hr
Flow @ 99.999 %	208 Nm3/hr
Flow @ 99.9995 %	167 Nm3/hr
Compressed Air Requirement	1344 CFM
Module	High Purity PSA system Only
Air Compressor And Drier	In your Scope / Quoted Separately

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Product	Description	Make	Qty
nGen Nitrogen Generator	nGEN 05 P Nitrogen		1

Caption	Specification
Gas Generated	Nitrogen
Gas Pressure	upto 6.5 Barg(Higher options Available)
Flow @ 99.99 %	9 Nm3/hr
Flow @ 99.995 %	7 Nm3/hr
Flow @ 99.999 %	5 Nm3/hr
Flow @ 99.9995 %	4 Nm3/hr
Compressed Air Requirement	32 CFM
Module	High Purity PSA system Only
Air Compressor And Drier	In your Scope / Quoted Separately

Commercial Offer

S No	Item Code	Item Name	Description	HSN	Qty	Unit Price	GST(%)	Amount
1	nGEN 1300 PC	nGEN 1300 PC	nGen Nitrogen Generator	840510	1	2,70,00,000.00	18.00	2,70,00,000.00
2	nGEN 1600 PC	nGEN 1600 PC	nGen Nitrogen Generator	840510	1	3,00,00,000.00	18.00	3,00,00,000.00
3	nGEN 150 P	nGEN 150 P	nGen Nitrogen Generator	840510	1	1,01,13,000.00	18.00	1,01,13,000.00
4	nGEN 05 P	nGEN 05 P	nGen Nitrogen Generator	840510	1	7,38,000.00	18.00	7,38,000.00
							Sub Total	6,78,51,000.00
							GST(%) @18.00	1,22,13,180.00
							Grand Total ()	8,00,64,180.00

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Terms And Conditions

1.	EXCLUSIONS & CLIENTâS SCOPE	All erection / civil work, All Power Cables & Cabling out of battery limit of PSA Plant, Cable Trenches etc. Unloading / Shifting of Gas Plant at Clientâs site will be in Client Scope. Additional gas piping, cooling water (if required).
2.	SUPERVISION OF ERECTION & COMMISSIONING	There shall be no commissioning charges, however you shall arrange freeTo & Fro Air Ticket, lodging / boarding, and local conveyance for our Engineers during supervision of Installation of Gas Plant at site on free of cost basis. Our Engineers will be there for Supervision only. Any manpower / fitter required for installation work shall be provided by you.
3.	P&F	3% Extra.
4.	Freight	Extra at Actuals.
5.	Insurance	In your Scope.
6.	Payment Terms	50% Advance with your Prestegious Purchase Order and Balance against PI.
7.	Delivery	12-16 Weeks from the PO and Advance. Rest can be confirmed for Availability.
8.	Warranty	We hereby confirm that, our equipment shall have Supplierâs warranty for a period of 12 months from the date of Commissioning or 18 months from date of Dispatch whichever is earlier.
9.	GST	As per the Commercial Table.
10.	Remarks	Nitrogen Storage needs to be discussed.

Hope you will find our offer in line with your requirement.

Should you require any further clarification/information, please feel free to contact us.

Thanking you, and assuring you of our best attention at all times.

Very Yours truly,

For Absstem Industrial Solutions Pvt. Ltd.

Amit Sahni

Chief Executive Officer

9582499594

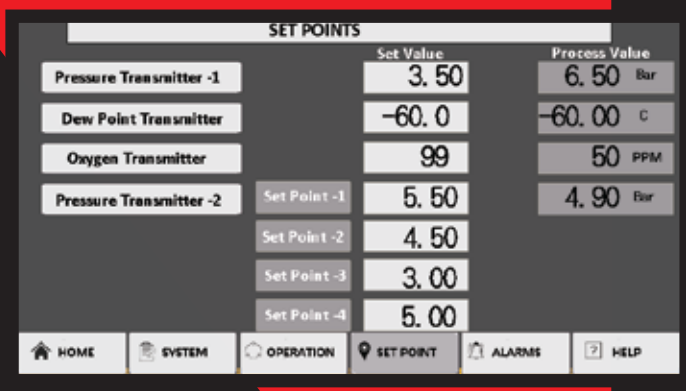
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THE EYE-TOUCH SCREEN

Real-time Data Visualization

Running Status
Live Storage Pressure
Running Hours
Service Due
Storage Gas Purity (% / PPM)
Flow
Consumption
Live Inlet Pressure





Adjustable Settings

Inlet Pressure On/Off Interlock
Dew Point Interlock
Storage Max Pressure Limit
Storage Pressure Alarm Control Adjust Point
Purity Adjust Point
Idle Mode
Auto Start Point

Alarm Systems

Purity Low Alarm
Dew Point Alarm
Other Customizable Alarms





Pop up Notifications

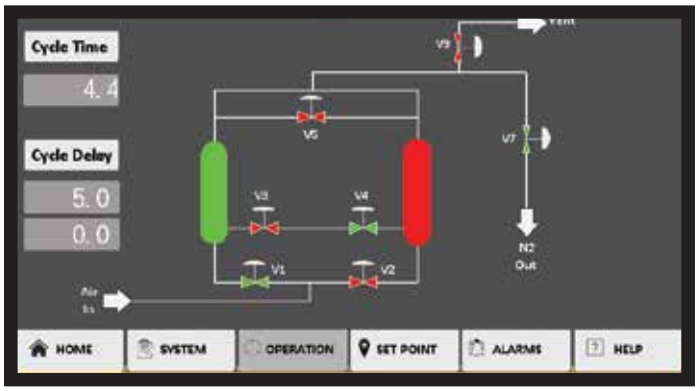
Low Inlet Air Pressure
Low Storage Pressure
Service Due Date

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absstem INDUSTRIAL

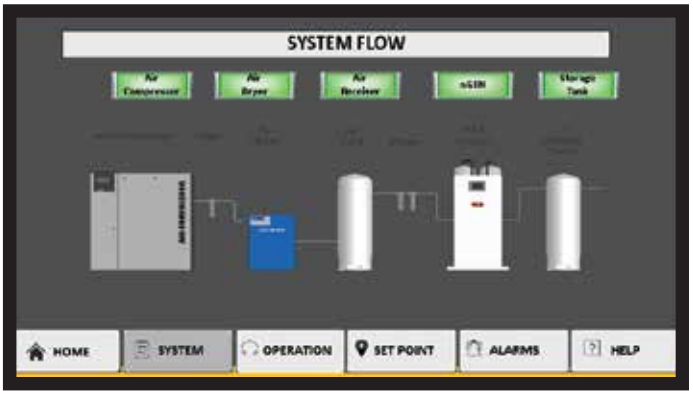
THE EYE-TOUCH SCREEN

Operation Screen



Live Valve Operation Status
Detailed Information

System Screen



Visual Diagram
External Storage / Communication

TECHNICAL SPECIFICATIONS

Display Section	Display Device	TFT Color LCD
	Screen Size	7" Wide Screen
	Resolution	WVGA: 800 × 480 dots
	Size	152.40 (6.00) (W) × 91.44 (3.60) (H) mm (inch)
	Backlight Life	Approx. 50000 h (Operating ambient temp: 25 °C, display intensity: 50%
Built-in Interface	RS-232	1 channel, Transmission Speed: 115200, 57600, 38400, 19200, 9600, 4800 bps Connector Shape: D-sub 9-pin (male)
	RS-422/485	1 channel, Transmission Speed: 115200, 57600, 38400, 19200, 9600, 4800 bps Connector Shape: D-sub 9-pin (female) Terminating Resistor: 330 Ω, 100 Ω, OPEN
	Ethernet	1 Channel Data Transfer Method: 100BASE-TX, 10BASE-T Connector Shape: RJ45 (modular jack) AUTO MDI/MDI-X
	USB (Host)	USB1.1 (Full-Speed 12 Mbps), Connector Shape: USB Mini-B
	SD card	1 channel, SDHC-compliant (max. 32 GB)
Others	Battery Life	GT11-50BAT Lithium Battery
	Dimensions	189 (7.44) (W) × 142 (5.59) (H) × 48 (1.89) (D) mm (inch)
	Weight	0.7 (1.5) kg (lb)

Additional Features

- Customizable User-friendly Interface
- Password Protection for secure access
- Data Logging for analysis and trend tracking
- Remote Access Capabilities
- DCS and SCADA-compatible
- Online Monitoring

Benefits

- Improved Operator-free Operation
- Enhanced System Monitoring and Control
- Reduced Downtime and Maintenance Costs
- Increased Safety and Process Stability
- Best Specific Energy as per Market

Note: This document provides a general overview. Specific Features and Functionalities may vary depending on your plant configuration and specific model of HMI is used.